2025 MIT Science Bowl High School Invitational

Round 11

TOSS-UP

1) ENERGY – *Short Answer* Researchers at MIT's Picower Institute have been studying the glymphatic system, a system similar to the function of the lymphatic system for circulating cerebrospinal fluid. In addition to neurons, the glymphatic system relies on what general types of cells to produce and circulate the CSF?

ANSWER: EPENDYMAL CELLS

BONUS

1) ENERGY – *Short Answer* Researchers at the Broad Institute are characterizing the functions of noncoding sequences in the genome. Identify all of the following three statements that are true about noncoding sequences: 1) lncRNAs are classified as noncoding; 2) Eukaryotes often have a larger part of their genome that is coding compared to prokaryotes; 3) snoRNAs are found in the nucleolus.

ANSWER: 1 AND 3

TOSS-UP

2) MATH – *Multiple Choice* Which of the following is closest to the limit as x approaches 0 of $e^x/\sin(e^x)$ (read: *the fraction with numerator e to the power of x and denominator sin of quantity e to the power of x*)?

W) 1
X) sin(1) (read: *sine one*)
Y) 1/sin(1) (read: *one divided by the quantity sine one*)
Z) The limit does not exist

ANSWER: Y) 1/SIN(1) (READ: *ONE DIVIDED BY THE QUANTITY SINE ONE*)

BONUS

2) MATH – *Short Answer* Three digits are selected at random, independently from the range 0 to 9. What is the probability that they can be arranged to form a three-digit number less than 300 without a leading zero?

ANSWER: 61/125

TOSS-UP

3) EARTH AND SPACE – *Multiple Choice* Which of the following rocks would be most expected to be formed near the surface along a hotspot track?

W) Skarn
X) Eclogite ***[eck-luh-jite]***
Y) Blueschist ***[blue-shist]***
Z) Gneiss ***[nice]***

ANSWER: W) SKARN

BONUS

3) EARTH AND SPACE – *Multiple Choice* Alice is a volcanologist sampling lava from a pahoehoe lava flow. What should she expect its silica content and volatile content to be, respectively, relative to ash samples gathered from a Plinian eruption?

W) Higher, higher
X) Higher, lower
Y) Lower, higher
Z) Lower, lower

ANSWER: Y) LOWER, HIGHER

TOSS-UP

4) CHEMISTRY – *Multiple Choice* Which of the following molecules in Earth's atmosphere cannot absorb infrared radiation?

W) Ozone
X) Carbon dioxide
Y) Nitrogen
Z) Methane

ANSWER: Y) NITROGEN

BONUS

4) CHEMISTRY – *Multiple Choice* Which of the following lithium group 13 metal hydride salts is the strongest reducing agent in organic synthesis?

W) Lithium borohydride
X) Lithium aluminum hydride
Y) Lithium gallium hydride
Z) Lithium indium hydride

ANSWER: X) LITHIUM ALUMINUM HYDRIDE

TOSS-UP

5) BIOLOGY – *Short Answer* Auxins only promote cell elongation at low concentrations. This is because at concentrations higher than 0.1 millimolar, auxin induces the production of what growth inhibiting hormone?

ANSWER: ETHYLENE

BONUS

5) BIOLOGY – *Short Answer* Identify all of the following three actions that would be able to alleviate spastic muscle paralysis in skeletal muscle: 1) Cleaving SNARE proteins; 2) Adding sarin; 3) Adding atropine.

ANSWER: 1 ONLY

TOSS-UP

6) PHYSICS – *Multiple Choice* Amy plots the reversible isothermal step in a Carnot cycle on a P-V diagram with log-log axes. Which of the following best describes Amy's graph?

W) Increasing and linear
X) Decreasing and linear
Y) Increasing and nonlinear
Z) Decreasing and nonlinear

ANSWER: X) DECREASING AND LINEAR

BONUS

6) PHYSICS – *Short Answer* When a laser of intensity 100 milliwatts is pointed along the length of a rectangular box of gas, 80 milliwatts are detected from the opposite side. The length of the cuvette is then doubled, and more gas is pumped in to maintain constant pressure and temperature. When the same laser is shone through the length of the box, how many millwatts are detected from the opposite side?

ANSWER: 64

TOSS-UP

7) EARTH AND SPACE – *Short Answer* In which layer of Earth's atmosphere are nacreous ***[neigh-cree-iss]*** clouds most commonly observed?

ANSWER: STRATOSPHERE

BONUS

7) EARTH AND SPACE – *Short Answer* Identify all of the following three characteristics of hurricanes which, due to climate change, are currently increasing on a global scale: 1) Annual frequency; 2) Average wind speed; 3) Average amount of precipitation.

ANSWER: 2 AND 3

TOSS-UP

8) PHYSICS – *Multiple Choice* Steph skydives from a very tall cliff with a large square parachute that has negligible mass and area A. Assuming uniform quadratic drag, the total time that Steph spends in the air is proportional to A raised to what power?

W) -1
X) -0.5
Y) 0.5
Z) 1

ANSWER: Y) 0.5

BONUS

8) PHYSICS – *Short Answer* A Science Bowl trophy has a weight of 100 newtons. Aryan hangs the trophy from the ceiling by connecting it to two strings on either side of the trophy, such that each string hangs 1 degree below the horizontal. Rounded to one significant figure and expressed in newtons, what is the tension in each string?

ANSWER: 3000

TOSS-UP

9) MATH – *Short Answer* What is the length of the shortest possible path on the unit sphere centered at the origin between the points $(1, 0, 0)$ and $(0, 1, 0)$?

ANSWER: $\pi/2$

BONUS

9) MATH – *Short Answer* Nine MIT Science Bowl volunteers want to choose some people amongst themselves to moderate finals. Given that they must choose at least one and at most four volunteers, how many ways are there to choose the moderators for finals?

ANSWER: 255

TOSS-UP

10) CHEMISTRY – *Short Answer* In a graphene lattice, what is the average bond order of each carbon-carbon bond?

ANSWER: 1.5

BONUS

10) CHEMISTRY – *Short Answer* How many valence electrons in a gas-phase diatomic molecule of carbon monofluoride reside in antibonding orbitals?

ANSWER: 3

TOSS-UP

11) BIOLOGY – *Multiple Choice* Most of the flesh in an apple is dervived from which of the following structures?

W) Receptacle
X) Endosperm
Y) Petiole
Z) Ovary

ANSWER: W) RECEPTACLE

BONUS

11) BIOLOGY – *Short Answer* Identify all of the following three processes which azidothymidine (read: *uh-zid-o-THY-mi-deen*), an analogue of thymidine, would inhibit: 1) HIV replication; 2) Retrotransposon duplication; 3) Telomere extension.

ANSWER: ALL

TOSS-UP

12) ENERGY – *Short Answer* Researchers at the Hammond Lab have been studying hemocyanin, a respiratory pigment found in arthropods. What is the central metal ion in hemocyanin?

ANSWER: COPPER

BONUS

12) ENERGY – *Short Answer* Researchers from CSAIL supported by the DOE are working on making exascale computation more efficient. Rank the following three sequence implementations by increasing expected memory usage for exascale computations: 1) Static array; 2) Dynamic array; 3) Linked list.

ANSWER: 1, 2, 3

TOSS-UP

13) CHEMISTRY – *Multiple Choice* Celsian ***[SELL-see-in]*** is a feldspar mineral formed when barium 2+ cations substitute for which of the following similarly sized cations?

W) Sodium 1+
X) Potassium 1+
Y) Calcium 2+
Z) Aluminum 3+

ANSWER: X) POTASSIUM 1+

BONUS

13) CHEMISTRY – *Short Answer* The second-order decomposition of 1 mole of substance A has an initial half life of 5 seconds. How long, in seconds, will it take for 1 mole of substance A to decompose to 0.1 moles?

ANSWER: 45

TOSS-UP

14) PHYSICS – *Short Answer* The molar heat capacity at constant volume of an ideal gas is 4.5 times the ideal gas constant. What is the adiabatic index of this gas?

ANSWER: 11/9

BONUS

14) PHYSICS – *Short Answer* Phoebe is taking measurements of different values in a circuit and wants to produce a graph that is linear in time. Identify all of the following three values that are linear when plotted with time, assuming ideal components: 1) Energy stored in an inductor powered by a constant voltage source; 2) Electric field inside a capacitor supplied by a constant current source; 3) Voltage across an inductor supplied by a constant current source.

ANSWER: 2 AND 3

TOSS-UP

15) MATH – *Multiple Choice* Which of the following quantities would be best described by a random variable following a Poisson ***[PWAH-sun]*** distribution?

W) The number of raindrops falling on a roof in 1 second
X) The measured lengths of rulers coming off of the manufacturing line
Y) The amount of time that lightbulbs last before burning out
Z) The amount of time between successive trains at a train station

ANSWER: W) THE NUMBER OF RAINDROPS FALLING ON A ROOF IN 1 SECOND

BONUS

15) MATH – *Short Answer* The point $(3, 4, 8)$ is reflected over the line $x = y = z$. What are the coordinates of the new point?

ANSWER: $(7, 6, 2)$

TOSS-UP

16) ENERGY – *Multiple Choice* Researchers in the Freedman Group at MIT are using diamond-anvil cells to create exotic phases of chemical compounds. These exotic phases lie on the extremes of which of the following directions on a standard pressure-temperature phase diagram?

W) Left
X) Right
Y) Top
Z) Bottom

ANSWER: Y) TOP

BONUS

16) ENERGY – *Short Answer* Physicists at MIT supported by the DOE modified perovskite crystals to build a quantum computing device that makes use of the intrinsic magnetic moment of electrons. What class of technologies does this device belong to?

ANSWER: SPINTRONICS

TOSS-UP

17) EARTH AND SPACE – *Short Answer* A star is observed with peak wavelength 300 nanometers. To one significant figure, what is its temperature in Kelvin?

ANSWER: 10,000

BONUS

17) EARTH AND SPACE – *Short Answer* A black hole's mass is instantaneously decreased by a factor of 5. By what factor does its luminosity instantaneously increase?

ANSWER: 25

TOSS-UP

18) BIOLOGY – *Short Answer* In red blood cells, what cytoskeletal protein forms a mesh-like network beneath the plasma membrane, providing structural support and flexibility?

ANSWER: SPECTRIN

BONUS

18) BIOLOGY – *Short Answer* Identify all of the following three statements concerning adult hemoglobin that are true: 1) The iron is coordinated to an aspartate in the porphyrin ring; 2) The oxygen binding curve of hemoglobin is shifted left when in the T state; 3) It would pass through a gel filtration chromatography faster than myoglobin.

ANSWER: 3 ONLY

TOSS-UP

19) MATH – *Short Answer* What is the smallest positive integer n such that $2^n \cdot n$ (read: *2 to the power of n times n*) is a multiple of 32?

ANSWER: 4

BONUS

19) MATH – *Multiple Choice* Which of the following functions over the real line does NOT have an inverse?

W) $f(x) = \log_2(x)$ (read: *f of x equals log base 2 of x*)
X) $f(x) = x \cdot \sqrt{x^2}$ (read: *f of x equals x times the square root of quantity x squared*)
Y) $f(x) = x + \sin(x)$ (read: *f of x equals x plus the sine of x*)
Z) $f(x) = 1 + 1/x$ (read: *f of x equals one plus quantity one over x*)

ANSWER: Z) $f(x) = 1 + 1/x$ (READ: *F OF X EQUALS ONE PLUS QUANTITY ONE OVER X*)

TOSS-UP

20) CHEMISTRY – *Multiple Choice* Which of the following solvents could be used with lithium aluminum hydride for organic synthesis?

W) Ethanol
X) Ethyl acetate
Y) Diethyl ether
Z) Dimethyl formamide

ANSWER: Y) DIETHYL ETHER

BONUS

20) CHEMISTRY – *Short Answer* Rank the following three organic compounds by increasing chemical shift of their signal in C-13 NMR: 1) Ethane; 2) Methyl bromide; 3) Methylmagnesium bromide.

ANSWER: 3, 1, 2

TOSS-UP

21) PHYSICS – *Multiple Choice* Quantum tunneling does not play a significant role in which of the following processes?

W) Pair production
X) Nuclear fusion
Y) Pyramidal inversion
Z) Alpha decay

ANSWER: W) PAIR PRODUCTION

BONUS

21) PHYSICS – *Short Answer* While the flow rate through an artery remains constant, plaque buildup causes its effective radius to decrease by 1 percent. To the nearest percent, by how much must the pressure difference across the artery increase to maintain the same flow rate?

ANSWER: 4 (ACCEPT: 4%)

TOSS-UP

22) ENERGY – *Short Answer* Scientists at the Broad Institute computed the cumulative distribution function of a probability distribution to find a decision boundary for binary classification. What value is the cumulative distribution function equal to at the decision boundary?

ANSWER: 0.5 (ACCEPT: 1/2)

BONUS

22) ENERGY – *Short Answer* The McGuire group is studying the vibrations of interstellar molecules such as cyanopentaacetylene ***[sigh-uh-no-pen-tuh-uh-set-uhl-een]***, a linear molecule containing 13 atoms. How many vibrational modes does cyanopentaacetylene have?

ANSWER: 34

TOSS-UP

23) BIOLOGY – *Multiple Choice* Daniel is prescribed a new drug called MIT, a zymogen (read: *ZY-muh-jen*) which mimics the activity of enterokinase and is cleaved in the small intestine. Which of the following enzymes would have its concentration directly increased as a result of administering this drug?

W) Pepsin
X) Phospholipase ***[foss-foe-lie-pays]***
Y) Trypsin
Z) Carboxypeptidase

ANSWER: Y) TRYPSIN

BONUS

23) BIOLOGY – *Multiple Choice* An XY baby with malfunctioning Sertoli cells cannot produce anti-Mullerian hormone. Which of the following phenotypes would result?

W) Development of only a female reproductive tract.
X) Development of only a male reproductive tract.
Y) Development of both a female and a male reproductive tract.
Z) Development of no reproductive tract

ANSWER: Y) DEVELOPMENT OF BOTH A FEMALE AND A MALE REPRODUCTIVE TRACT.

TOSS-UP

24) EARTH AND SPACE – *Short Answer* Identify all of the following three structures which, by mass, are primarily composed of water ice: 1) The rings of Jupiter; 2) The rings of Saturn; 3) The lakes of Titan.

ANSWER: 2 ONLY

BONUS

24) EARTH AND SPACE – *Multiple Choice* Which of the following is closest to the ratio between the volumes of Olympus Mons on Mars and Mauna Loa on Earth?

W) 1
X) 10
Y) 100
Z) 1000

ANSWER: Y) 100

TOSS-UP

25) CHEMISTRY – *Multiple Choice* Which of the following transition metals would be most likely to form a square planar complex when coordinated by four monodentate ligands?

W) Iron
X) Zinc
Y) Silver
Z) Platinum

ANSWER: Z) PLATINUM

BONUS

25) CHEMISTRY – *Multiple Choice* Jennifer constructs a galvanic cell using a standard hydrogen electrode and a zinc electrode submerged in a saturated solution of zinc fluoride. Which of the following changes would increase the voltage of Jennifer's galvanic cell?

W) Adding sodium fluoride to the zinc cell
X) Adding zinc sulfate to the zinc cell
Y) Increasing the size of the zinc electrode
Z) Adding potassium hydroxide to the standard hydrogen electrode

ANSWER: W) ADDING SODIUM FLUORIDE TO THE ZINC CELL